

Investment Strategy Challenge 2026 · Proposal for NKE Private Wealth (Chak family)

Asian Digital Transformation Autocallable Note

*10.25% a year from Asia's semiconductor build-out, with at least
70% of capital back whatever happens*

USD 100mn · 5-year note issued by Natixis · 10.25% p.a. memory coupon · 70% barrier + embedded
70% put · Trade date 17 Sep 2026

Agenda

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Client & market

Who the Chak family is, what they need, and why Asia's digital build-out now

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Investment thesis & strategy

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Risks & hedging

Investor risks, mitigants, and how the Natixis desk hedges

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Executive summary

One note that turns the family's operating expertise into double-digit income with a capped downside

10.25%

Memory coupon p.a.

2.56% each quarter the basket is $\geq$ 100%; missed coupons are paid later when it recovers

70%

Minimum repayment

100% back unless the basket ends below 70%; then the embedded put still pays 70%. Maximum loss 30% (plus Natixis credit)

49%

Called at first review

Risk-neutral. 79% called within 5 years; expected life 2.22 years

We recommend a **USD 100mn, 5-year Natixis buffered autocallable note** on a basket of five Bloomberg-listed Asian ETFs: Global X Asia Semiconductor (3119 HK) 35%, iShares MSCI South Korea (EWY) 20%, iShares MSCI Taiwan (EWT) 15%, Global X Japan Semiconductor (2644 JP) 15% and Cathay Taiwan ESG High Dividend (00878 TT) 15%. It gives the family the AI hardware layer (chips, memory and chip-making equipment) that complements, rather than duplicates, the infrastructure, power and property they already own. **The family earns 10.25% a year in the 79% of paths where the basket gets back to its starting level, gets 100% back if it ends anywhere above 70%, and never less than 70% (16% of paths, risk-neutral). Fair value is 97.1%, so issuing at par leaves Natixis 2.9% for hedging and margin.**

The family wants growth without putting the legacy at risk

Client diagnosis: third-generation dynasty, succession done, now allocating for legacy

Who they are

- Single-family office: NKE Private Wealth
- Net worth > USD 2bn; mandate USD 100mn (~5%)
- Businesses: tech infrastructure, renewable power plants, real estate in Japan and Greater China

What they want

- Capital appreciation for future generations
- Meaningful exposure to Asia's digital transformation
- To bridge traditional Asian sectors with emerging tech they know

What they can bear

- Moderate risk: preserve the legacy first
- 5-year horizon; coupons can be reinvested
- Accepts limited liquidity on 5% of wealth
- Volatile backdrop: loss must be capped

Client need	Design answer in our note
Capital appreciation	10.25% p.a. coupon, above the 5.69% 5Y USD funding rate, in every path that autocalls; proceeds roll into the next note
Exposure to Asian digital transformation	Semiconductor-heavy basket: the AI hardware layer behind the family's tech-infrastructure business (slide 5)
Generational preservation	70% barrier plus embedded 70% put: maximum loss 30% at maturity; diversified basket (not worst-of); BPCE-backed issuer
Volatile markets	Memory coupon: a missed coupon is paid when the basket recovers; a fall of up to 30% at maturity costs nothing

Why now: Asia builds the hardware of the AI economy

Investment thesis: structural growth themes, and high rates and volatility that pay for a coupon with a floor

1 Taiwan: the AI foundry

TSMC and its supply chain make most of the world's advanced AI chips; Taiwan's index is dominated by semiconductors.

2 Korea: memory for AI

Samsung and SK hynix supply the high-bandwidth memory every AI accelerator needs; together ~44% of EWY.

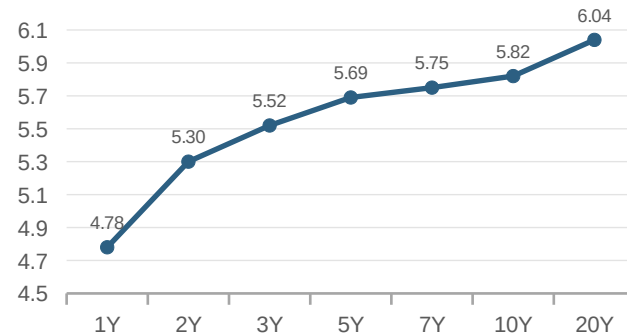
3 Japan: chip-making tools

Tokyo Electron, Advantest and peers make the equipment and testers chip factories need, alongside TSE governance reforms.

4 ESG sleeve

00878 adds ESG-screened Taiwan leaders (MSCI ESG methodology) and a lower-volatility, high-dividend anchor to the basket.

USD funding curve (case grid, %)



What the curve means for structuring

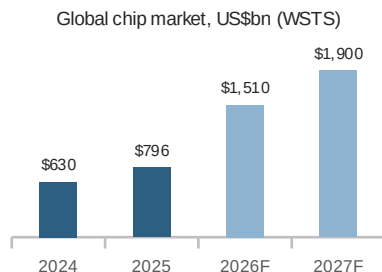
High USD rates make the principal cheap to fund (5Y zero-coupon 75.8%). Asian volatility (basket ~26%) makes the basket likely to revisit 100%, which pays the 10.25% coupon, and the embedded 70% put caps the loss at 30%.

Thesis is qualitative; index-level data to be refreshed on Bloomberg. Funding grid: Investment Strategy Challenge 2026 rules.

Investment thesis: Monopolizing the AI Supply Chain via Structured Overlays

What we own and why we wrap it: structural demand for Asian chips, memory and tools, held through a market of high rates, record volatility and concentrated, full prices

1 Asia controls AI's hardware bottlenecks



Demand is structural

Hyperscaler capex ~US\$725bn in 2026 (+77%); the chip market nearly doubles to ~US\$1.5trn.

The moat runs through Asia

8 of the world's 10 largest companies depend on TSMC (J.P. Morgan).

...but the cycle is maturing

DRAM price gains slowing: +48% in Q2, ~+13% expected in Q4 2026.

~70%

Taiwan makes the chips: TSMC share of global foundry revenue; TSMC >40% of TAIEX

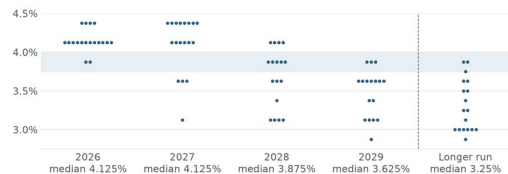
83%

Korea makes the memory: SK hynix + Samsung share of HBM, the memory every AI accelerator needs

~55%

Japan makes the tools: Advantest share of chip test equipment; with Tokyo Electron ≈ 20% of the Nikkei

2 2026 punishes bonds and buy-and-hold



Rates stay high

Fed median 4.125% to end-2027, cuts only from 2028; 12 dots see 4.375%.

FOMC dot plot, 16 Sep 2026 (shaded: today's 3.75–4.00%)

–4.0%

Long bonds carry rate risk: a 5-year bond loses ~4.0% per 1% rise.

→ Redeems from year 1; USD 75.8 today buys USD 100 in 5 years, so a 70% floor fits

91.2

Volatility is at records: VKOSPI's all-time high; KOSPI fell ~25% from June to August 2026.

→ Coupons pay on recovery, no timing needed; rich option premium funds them

~92%

Concentrated and fully priced: of advanced chips come from Taiwan; Nikkei CAPE 38.6×

→ Diversified five-ETF basket; capping upside at the coupon costs little

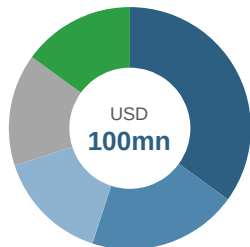
Whichever AI company wins, it needs Asia's chips, memory and tools. In 2026's market, the way to own them is an autocall with a 30% buffer and a 70% floor.

Sources: WSTS; company capex guidance; Counterpoint (foundry, HBM); TrendForce (DRAM); industry research (test equipment); J.P. Morgan AWM 2026 Outlook; Federal Reserve SEP (16 Sep 2026, dots read from chart); Herald / SBS (VKOSPI); FN News, BIT Research (KOSPI); Sibilis (CAPE); case USD funding grid. Full list: deck/thesis_sources.md.

Strategy: our five-ETF basket sits at the low-risk edge of every possible mix

USD 100mn across five Bloomberg-listed ETFs; our weights (red star) deliver ~26% expected return for only ~17% volatility

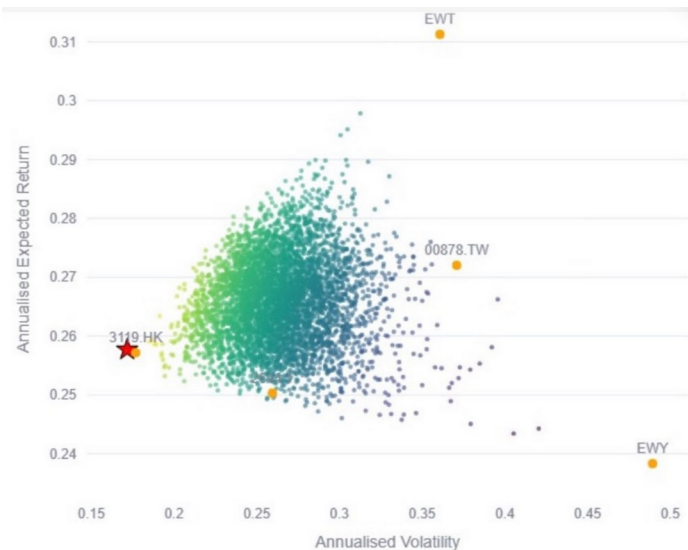
Allocation



Half in the semiconductor core
3119 HK + 2644 JP = 50%; Korea adds memory, Taiwan the foundry, 00878 the ESG income.

■	3119 HK	Pan-Asian chip leaders	35%
■	EWY US	Korea: memory for AI	20%
■	EWT US	Taiwan: the AI foundry	15%
■	2644 JP	Japan: chip-making tools	15%
■	00878 TT	ESG dividend anchor	15%

Risk vs return: single ETFs and random mixes



How to read it

Orange dots: each ETF on its own (2644 is the dot near 0.26, 0.25).

Red star: our basket (35 / 20 / 15 / 15 / 15).

Small dots: thousands of random weightings of the five ETFs; lighter = more return per unit of risk.

Axes: annualised volatility (risk) and expected return.

Takeaways

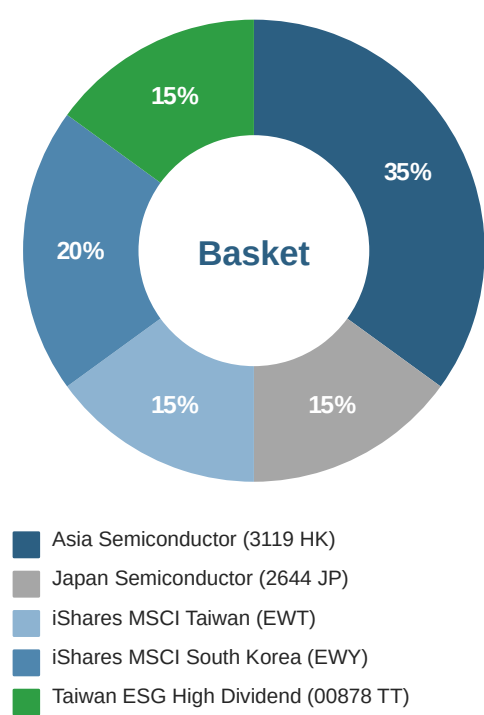
- Our basket (red star): ~26% return at ~17% volatility, the lowest risk of all the mixes shown.
- Most random mixes need 22–32% volatility for the same 25–28% return: our weights get it with less risk.
- Less risk than any single ETF, about a third of EWY's (~49%), while giving up only a few points versus EWT's ~31%.
- Why: 3119 HK (35%) anchors the basket; the other four move differently enough to pull risk down.

Our basket (red star): ~26% return for ~17% volatility, less risk than any single ETF or random mix

Risk/return chart: team analysis app (annualised from historical data; red star = our basket at 35/20/15/15/15; values read from the chart). Liquidity: 3119 HK is borderline against the USD 5M/day rule, verify on Bloomberg.

Underlying: five Asian ETFs across the AI-hardware chain

Bespoke weighted basket, USD payout (quanto on the HKD, JPY and TWD lines); all Bloomberg-listed



ETF (Bloomberg)	Wt	Role in the basket	Vol*
Asia Semiconductor (3119 HK)	35%	Pan-Asian chip leaders (40 names; TSMC, Samsung, SK hynix the largest): the core AI-hardware exposure	32%
Japan Semiconductor (2644 JP)	15%	Japanese semiconductor stocks (FactSet Japan Semiconductor Index, e.g. Tokyo Electron, Advantest): the family's home market	38%
iShares MSCI Taiwan (EWT)	15%	Taiwan large caps (TSMC, MediaTek, Delta): AI foundry and server chain in the family's Greater China region	26%
iShares MSCI South Korea (EWY)	20%	Korea large caps (Samsung + SK hynix ~44%): high-bandwidth memory for AI	30%
Taiwan ESG High Dividend (00878 TT)	15%	ESG-screened Taiwan high-dividend leaders (Quanta, UMC, MediaTek): ESG sleeve and lower-volatility anchor	18%

Why a weighted basket, not worst-of: a shock to one ETF is cushioned by the others: basket vol ~26% vs 18%–38% alone. Overlap (TSMC, Samsung, SK hynix sit in several ETFs) keeps correlations high (0.45–0.85).

*Assumed 5Y vols. USD 5M/day ETF rule: EWY, EWT, 2644, 00878 pass; 3119 is borderline (~USD 4–5M/day), verify the 6-month average on Bloomberg. 00878 distributes ~7.6% a year, which lowers its price path.

Term sheet: 5-year buffered autocallable, 10.25% coupon

Why this design: a double-digit coupon, with the loss at maturity capped at 30%

Issuer	Natixis SA (BPCE Group)
Notional / currency	USD 100,000,000 · quanto USD
Trade / maturity	17 Sep 2026 / 17 Sep 2031 (5Y)
Underlying	Weighted basket of 5 Asian ETFs (slide 5)
Observation	Quarterly, 20 dates; autocall from Q4 (Year 1)
Coupon	10.25% p.a. (2.56% per quarter), paid if basket $\geq$ 100%
Memory	Missed coupons paid on the next date basket $\geq$ 100%
Autocall	Basket $\geq$ 100% from Q4: 100% + coupons due, note ends
At maturity	100% if basket $\geq$ 70%; otherwise 70% (embedded 70% put)
Issue price / fair value	100% / 97.1%

Autocall design (same basket, issued at 98%)	Fair cpn	Loss risk	Max loss
Classic fixed coupon, 70% barrier	7.4%	15.9%	Up to 100%
Snowball, 65% barrier	13.7%	14.3%	Up to 100%
Memory coupon, 70% barrier, no put	13.5%	15.9%	Up to 100%
Memory coupon, 100% protected	7.8%	0.0%	None
Memory coupon, 70% barrier + 70% put (ours)	11.1%	15.9%	30%

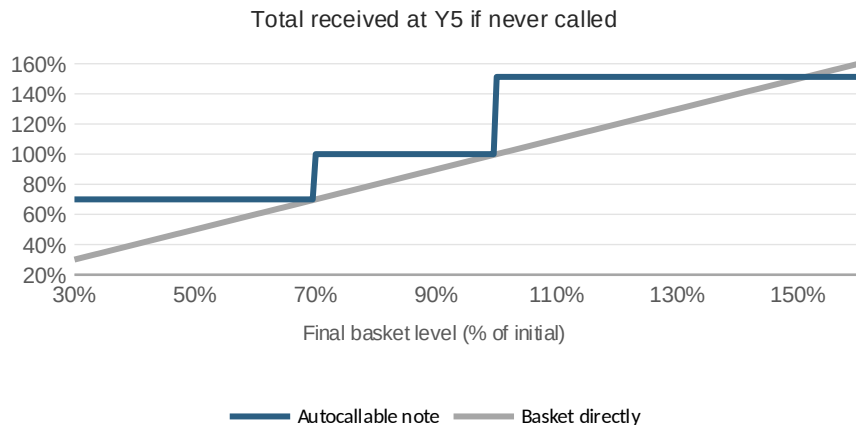
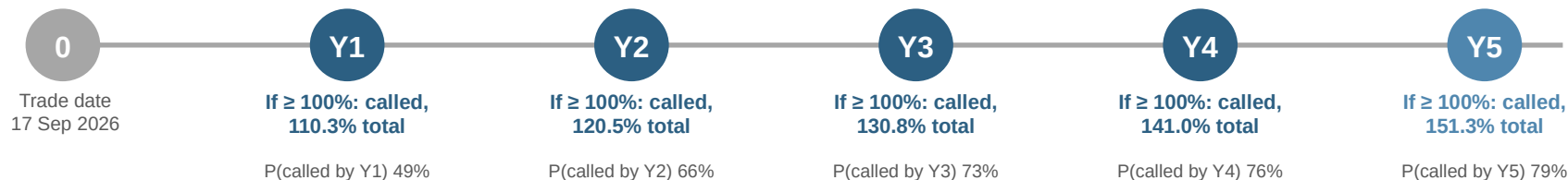
Design call: The draft's 8.5% fixed coupon is worth 100.3%, so Natixis cannot fund it. Full protection pays only 7.8%. Without a put the coupon is 13.5%, but a crash can wipe out the capital. The embedded 70% put costs ~2.4pts of coupon and caps the loss at 30%: fair 11.1%, we offer 10.25%.

Monte Carlo, 200k paths. Vols, dividends and correlations are assumptions to refresh with Bloomberg data as of the 17 Sep 2026 trade date; discounting uses the USD funding grid. Loss risk = probability the note returns less than 100% at maturity (risk-neutral).

How it pays: one question every quarter, three outcomes

Is the basket at or above 100% of its starting level? (checked quarterly; one date per year shown)

Q1–Q3: coupon only (2.56% if basket $\geq 100\%$). From Q4: coupon plus autocall. Missed coupons are always caught up.



A

Autocalled

Basket $\geq 100\%$ on any date from Q4: 100% plus every coupon to date. The note ends.

79%
risk-neutral

B

Not called, basket $\geq 70\%$ at Y5

The 30% buffer absorbs the fall: 100% back, plus any early coupons.

5%
risk-neutral

C

Not called, basket $< 70\%$ at Y5

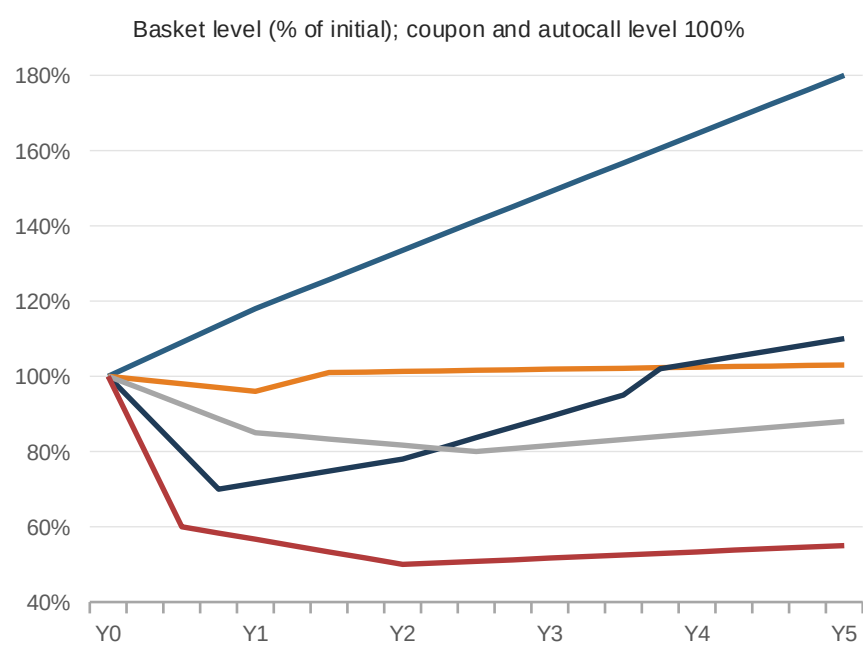
The embedded put pays 70% back. Maximum loss 30%.

16%
risk-neutral

Probabilities are risk-neutral (drift = funding rate – dividends); real-world outcomes are on slide 9. The desk's replication (bond, coupon digitals, 70% put pair) is in Appendix B.

Scenario analysis: five paths for Asian markets

Hypothetical quarterly basket paths and what the USD 100mn note returns in each



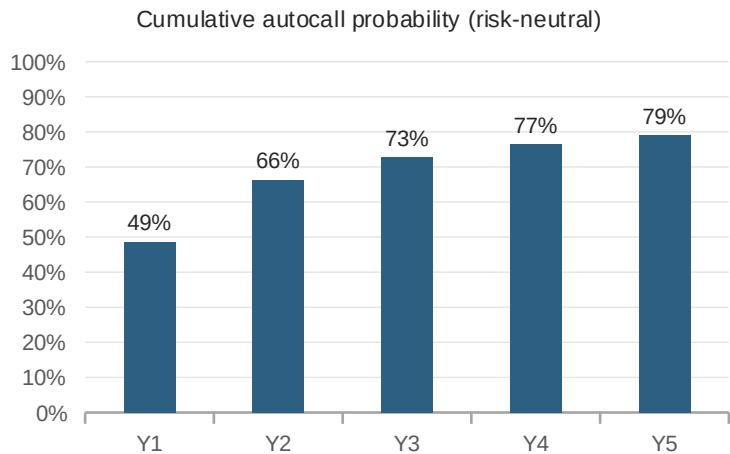
Scenario	Outcome	Coupons	USD mn	IRR	Basket
AI capex boom	Called Q4 (1y)	10.3%	110.3	10.7%	—
Range-bound Asia	Called Q6 (1.5y)	15.4%	115.4	10.0%	—
2022-style tech drawdown, slow recovery	Called Q15 (3.75y)	38.4%	138.4	9.1%	—
Lost decade (sideways below par)	100% back at Y5	0.0%	100.0	0.0%	-12%
Geopolitical shock, no recovery	70% back at Y5 (put)	0.0%	70.0	-6.9%	-45%

Reading: the note earns ~9.1%–10.7% a year whenever the basket gets back to 100%, even after a 30% drawdown: memory pays all 15 missed coupons at Q15. In a lost decade (basket –12%) the buffer returns 100%; in a lasting shock (basket –45%) the embedded put returns 70%.

Paths are illustrative and not forecasts. 'Basket' = direct basket return at Y5 when the note is not called. Historical back-test to run on Bloomberg history (appendix).

Simulation back-test: most paths end early with full coupons

Monte Carlo outcome distribution: risk-neutral and under three real-world equity return assumptions



Metric	Risk-neutral	0% equity p.a.	4% equity p.a.	8% equity p.a.
Called within 5Y	79%	68%	77%	84%
Expected life (yrs)	2.2	2.6	2.3	2.0
Mean IRR	7.1%	5.2%	6.7%	7.9%
Median IRR	10.4%	10.1%	10.3%	10.4%
Avg coupons received	12.3%	10.9%	12.0%	12.8%
Zero return (100% back)	3%	3%	3%	3%
70% back (put pays)	16%	27%	18%	11%

49%

called at the first review
(110.3% total)

16%

chance of 70% back:
loss capped at 30%

70%

worst case at maturity, in
every path

Monte Carlo, 200k paths. Vols, dividends and correlations are assumptions to refresh with Bloomberg data as of the 17 Sep 2026 trade date; discounting uses the USD funding grid. IRR uses actual quarterly coupon dates.

Risks for the family, and how Natixis hedges its side

Transparent disclosure plus a desk hedging plan for every exposure

Risk	What could happen	Mitigant
Capital loss	Basket below 70% at Y5: 70% back, a 30% loss (~16% of paths)	Embedded 70% put caps the loss; diversified basket; only 5% of wealth
Capped upside	Basket +80% still pays 10.25% p.a. and is called at Y1	Rest of family wealth keeps direct equity exposure
ETF structure	00878 pays ~7.6% a year out, lowering its price path; 3119 trading is thin; ETF tracking error	00878 held at 15%; verify volumes; Natixis can reference total-return versions
Issuer credit	Repayment and the 70% floor depend on Natixis / BPCE	BPCE senior rating; optional collateralised wrapper
Liquidity / MTM	Sold before maturity, the price can be below 100%	Natixis daily indicative price; hold to call or maturity

Pricing sensitivity (fair coupon): base 11.1% · vol ±3pts 10.3%–11.9% · corr ±0.15 10.6%–11.6% · equity drift –1/–2% 12.0%–13.2%. The 10.25% offer stays fundable in every case.

Coupon digitals

Natixis owes basket digitals struck at 100%: replicated with tight basket call spreads, delta-hedged with ETF units, constituent stocks and index futures as proxies; listed options on EWY/EWT

Vol & correlation

Coupon value moves with basket vol and correlation (fair 10.3%–11.9% for vol ±3pts): managed with listed index options and correlation trades

Quanto FX

EWT and EWY already trade in USD; 3119 (HKD, pegged), 2644 (JPY) and 00878 (TWD) need FX forwards and a quanto adjustment

Put pair & funding

Natixis is long the client's 70% barrier put and short the embedded 70% put: a 30% digital put, hedged with basket put spreads. Principal funded at 5.69%

Appendix A: pricing assumptions and methodology

Inputs to refresh on Bloomberg at trade date (BVOL / OVDV for vols, CORR for correlations)

ETF	Weight	Vol	Div	3119 HK	2644 JP	EWT US	EWY US	00878 TT
3119 HK	35%	32%	1.0%	1.00	0.70	0.85	0.80	0.60
2644 JP	15%	38%	0.8%	0.70	1.00	0.60	0.55	0.45
EWT US	15%	26%	2.6%	0.85	0.60	1.00	0.70	0.75
EWY US	20%	30%	1.2%	0.80	0.55	0.70	1.00	0.55
00878 TT	15%	18%	7.6%	0.60	0.45	0.75	0.55	1.00

Model. Correlated GBM for each ETF, 200,000 paths, quarterly steps. Risk-neutral drift = USD funding (linear interpolation of case grid) – dividend yield. Quanto adjustment ignored (assumed hedged by the desk; to be added with FX-equity correlations).

Discounting. USD funding grid from the case rules; 5Y discount factor 0.758.

Coupon solve. Bisection on coupon so that PV = 98% (2% hedging cost and margin).

Historical back-test (to run). Monthly Bloomberg history since 2021 for all five ETFs (00878 and 3119 launched in 2020–21). Launch a hypothetical note every month, apply the same coupon, memory and autocall rules, and record coupons, life and IRR by launch year.

Variant of our note (70% barrier + put)	Fair cpn
Coupon barrier 100% (base)	11.1%
Coupon barrier 95%	10.2%
Coupon barrier 90%	9.4%
First autocall at Y2	10.7%
Equity drift –1%	12.0%
Equity drift –2%	13.2%
Vol +3pts / –3pts	11.9% / 10.3%

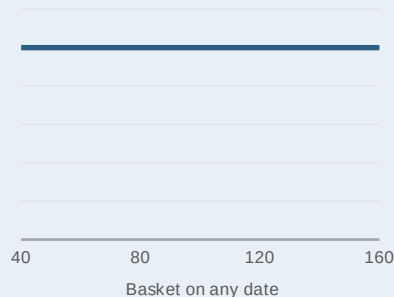
All variants priced at 98% with the same basket and inputs. Code: [pricing/scenario2_autocall_mc.py](#)

Appendix B: how Natixis replicates the note

Replication used by the Natixis desk to price and hedge (values in % of notional, at trade date)

Principal

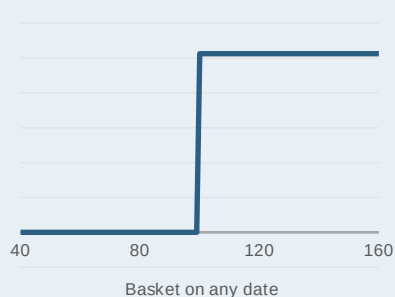
100% repaid at autocall or maturity.
Worth 89.4%



+

Coupon digitals

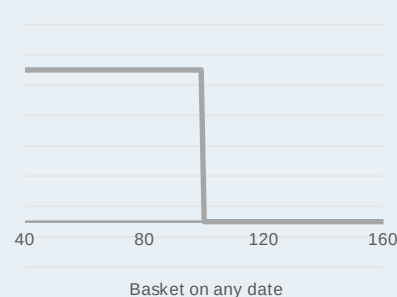
2.56% plus missed coupons on each date $\geq 100\%$. Worth 11.3% (net of autocall)



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Autocall (Natixis)

From Q4, basket $\geq 100\%$: remaining coupons cancelled, note ends. Keeps the coupon affordable



-

70% put pair

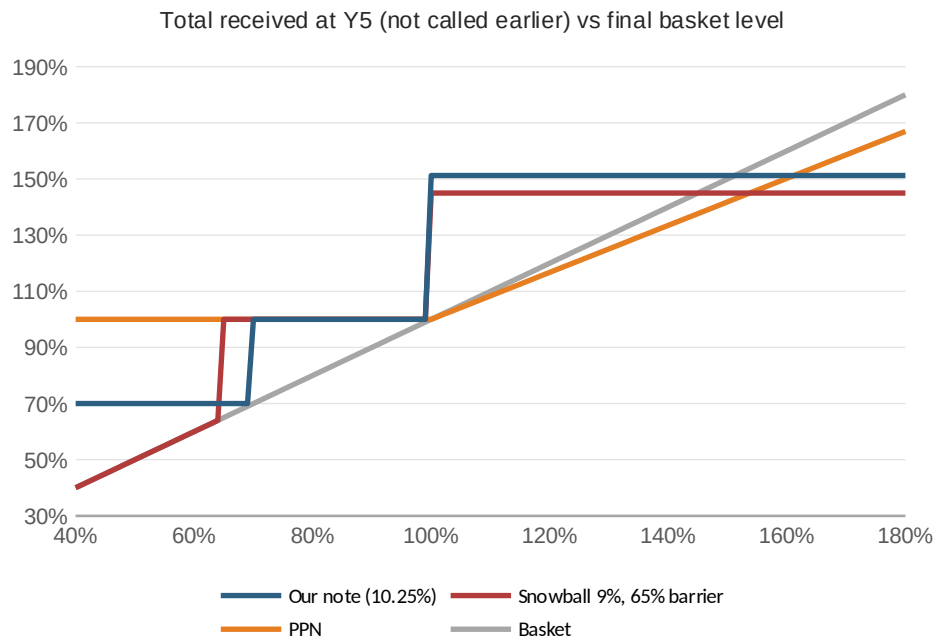
Client sells a 70% barrier put (6.3%); the note embeds a 70% put back (2.7%).
Net: -30% only if $< 70\%$ at Y5



Fair value 97.1% = principal 89.4% + coupons 11.3% – put pair 3.6%. Issued at 100%, leaving 2.9% for hedging and margin. The short barrier put funds the higher coupon; the embedded put stops the loss at 30%.

Appendix C: alternatives for the investment committee

Same basket: full protection, a higher-coupon at-risk snowball, and a pure participation note



Option	Return	Loss risk
Buffered autocallable (ours)	10.25% p.a. memory coupon	16% (max 30%)
100% protected autocallable	~7.8% fair coupon	0%
Snowball, 65% barrier	9% p.a., paid at call	14.3%
Principal-protected note	84% of basket gain at Y5	0%

Trade-off: The snowball pays 9% but loses capital in ~14% of paths. Full protection pays only ~8%. The PPN pays nothing unless the basket ends above 100% after 5 years. Our note pays 10.25% and caps the loss at 30%, with money usually back within ~2 years.

Monte Carlo, 200k paths. Vols, dividends and correlations are assumptions to refresh with Bloomberg data as of the 17 Sep 2026 trade date; discounting uses the USD funding grid.